Lesson 6.10 **Duration: 40 minutes**

Ethics in Technology

Learning Objectives

- Identify ethical dilemmas arising from emerging technologies
- Apply ethical frameworks to evaluate technology decisions
- Propose responsible technology development practices

Engage (5-7 minutes)

Present a dilemma: an AI system can predict crimes before they happen (Minority Report scenario). Is this ethical? Who decides what constitutes a "crime"? Students debate: safety versus privacy, efficiency versus freedom, progress versus caution. Establish that technology is never neutral—it reflects the values of its creators.

Explore (10-12 minutes)

Students analyze four case studies: facial recognition bias (higher error rates for people of color), algorithmic lending discrimination (denying loans based on zip code), deepfakes (fabricated videos), and social media addiction (algorithmic amplification of outrage). For each, they identify: the technology, the harm, who is affected, and potential solutions.

Explain (10-12 minutes)

Technology ethics examines moral principles guiding the design, development, and deployment of technological systems. Key issues include: privacy (data collection and surveillance), bias (AI systems perpetuating discrimination), transparency (algorithmic decision-making opacity), accountability (who is responsible when technology causes harm), and autonomy (technology manipulating behavior). Frameworks for analysis include utilitarianism (greatest good for greatest number), deontology (duty-based rules), and virtue ethics (character of the decision-maker). Ethical technology requires diverse teams, inclusive design, and proactive regulation.

Elaborate (8-10 minutes)

Deep-dive into specific ethical challenges: AI hiring tools that discriminate against women (Amazon's case), predictive policing that reinforces racial bias, social media algorithms that prioritize engagement over truth, and genetic technology that raises questions about eugenics. Discuss the role of regulation: GDPR in Europe, proposed AI regulations, and the tension between innovation and protection. Examine corporate responsibility: tech companies' ethical boards, employee activism, and the "move fast and break things" mentality versus responsible development.

Evaluate (5-8 minutes)

Students participate in a structured ethical analysis: given a technology scenario (e.g., AI grading students, facial recognition in schools, predictive analytics for student dropout), they must identify stakeholders, evaluate using two ethical frameworks, propose a balanced solution, and defend their position. Written reflection: what responsibility do technologists have for the impact of their creations?